

ElixirML: Predicting Chirality-Driven Differential Activity in Longevity-Associated Drug Targets

A Machine Learning Pipeline Using Spectral Chirality Fingerprints
and Target-Conditioned Models

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Abstract

We present ElixirML, a machine learning pipeline for predicting stereoselective differential activity in longevity-associated drug targets. Using spectral chirality fingerprints derived from dual-helix graph Laplacian decomposition, we train three complementary models on 170 enantiomer pairs spanning seven longevity-relevant pathways (p53, mTOR, PI3K, SIRT, proteasome, telomerase, AMPK). Our key finding is that chirality fingerprint divergence predicts fold-change *only within specific pathways*—p53-MDM2 shows strong correlation ($\rho_{\text{Spearman}} = 0.406$, $p = 0.001$) while mTOR and proteasome show none ($\rho < 0.1$). A target-conditioned global model achieves $\rho_{\text{Spearman}} = 0.549$ ($p < 10^{-14}$) via leave-one-out cross-validation, demonstrating that molecular identity plus target context is required for fold-change prediction. We identify 36 “golden” drug-like candidates with extreme chirality-driven potency differences ($>10\times$ fold-change, divergence > 0.7 , Lipinski-compliant), led by ChEMBL4759169, a PI3K- δ inhibitor exhibiting $3,551\times$ enantioselectivity. SHAP analysis reveals pathway-specific prediction mechanisms: spectral fingerprint dimensions drive regression while divergence dominates classification. These results establish that chirality-aware molecular fingerprints, when conditioned on biological target context, can identify longevity-relevant compounds where stereochemistry is the decisive factor in therapeutic activity.

Keywords: chirality, enantiomers, longevity, drug discovery, spectral fingerprints, machine learning, SHAP, PI3K, mTOR, p53

1 Introduction

Stereochemistry is among the most consequential molecular properties in drug design. Enantiomers—mirror-image molecules differing only in the spatial arrangement of atoms around one or more chiral centers—can exhibit dramatically different pharmacological profiles. The thalidomide tragedy, where (R)-thalidomide was therapeutic while (S)-thalidomide was teratogenic, remains the paradigmatic example [1]. In longevity research, where the therapeutic targets include kinases (mTOR, PI3K), deacetylases (sirtuins), tumor suppressors (p53), and the proteasome, the role of molecular chirality is poorly characterized.

Prior work in this project (Atlas v8) established that standalone chirality fingerprint divergence—a measure of how different two enantiomers appear to a spectral chirality detector—*cannot* predict fold-change magnitude across all targets ($\rho_{\text{Spearman}} = -0.008$, $n = 1,998$). This negative result is scientifically important: it means that knowing how “chirally different” two

molecules are tells you nothing about how differently they will behave biologically, unless you also know what biological target they are acting on.

ElixirML tests the hypothesis that **chirality fingerprints conditioned on target identity can predict fold-change**, even though the fingerprints alone cannot. We construct three models:

1. A **fold-change regressor** predicting $\log_{10}(\text{FC})$ from differential fingerprint pairs plus pathway context,
2. A **HIGH classifier** identifying pairs with extreme chirality-driven potency differences, and
3. A **global-style model** using the architecture from our 138K-record ChEMBL model, evaluated on longevity pairs via leave-one-out cross-validation.

1.1 Contributions

- Demonstration that chirality fingerprint–fold-change correlation is *pathway-specific* (strong in p53, absent in mTOR/proteasome)
- A target-conditioned model achieving $\rho_{\text{Spearman}} = 0.549$ on longevity pairs where unconditional models fail
- Identification of 36 golden drug-like candidates with extreme stereoselective potency
- SHAP-based mechanistic insight into which molecular features drive chirality-dependent activity in each pathway

2 Background

2.1 Spectral Chirality Fingerprints

The chirality fingerprint system used in this work is based on dual-helix spectral graph decomposition [2]. For a molecule with adjacency matrix $\mathbf{A}$, we construct two graph Laplacians:

$$\mathbf{L}_{\text{std}} = \mathbf{D} - \mathbf{A}_{\text{std}}, \quad \mathbf{L}_{\chi} = \mathbf{D}_{\chi} - \mathbf{A}_{\chi} \quad (1)$$

where $\mathbf{A}_{\chi}$ incorporates chirality-modulated edge weights via CIP-priority canonical ordering with cyclic shifts. The eigenvalue spectra of both Laplacians are computed:

$$\lambda_{\text{std}} = \text{eig}(\mathbf{L}_{\text{std}}), \quad \lambda_{\chi} = \text{eig}(\mathbf{L}_{\chi}) \quad (2)$$

The **differential fingerprint** uses only the chirality-sensitive components:

$$\mathbf{f}_{\text{diff}} = \text{proj}([\lambda_{\chi}^{\cos} - \lambda_{\text{std}}^{\cos}, \lambda_{\chi}^{\sin} - \lambda_{\text{std}}^{\sin}]) \in \mathbb{R}^{128} \quad (3)$$

where $\text{proj}(\cdot)$ is a deterministic random projection (seed 42) from raw dimension to 128 bits. For enantiomers, $\mathbf{f}_{\text{diff}}(R) \approx -\mathbf{f}_{\text{diff}}(S)$, producing cosine similarity ≈ -1.0 .

The **fingerprint divergence** between enantiomers A and B is:

$$\text{div}(A, B) = 1 - \text{sim}(\mathbf{f}_A, \mathbf{f}_B), \quad \text{sim}(\mathbf{f}_A, \mathbf{f}_B) = \frac{\mathbf{f}_A \cdot \mathbf{f}_B + 1}{2} \quad (4)$$

2.2 Longevity Target Pathways

We focus on seven pathways implicated in aging and longevity:

Table 1: Longevity-associated pathways and their biological roles.

Pathway	Primary Targets	n pairs	Longevity Mechanism
p53	MDM2, MDM2-p53 PPI	60	Tumor suppression, senescence
mTOR	mTOR kinase, PI3K α	59	Nutrient sensing, autophagy
Proteasome	β -subunits, 20S	35	Proteostasis, protein quality
PI3K	p110- δ , p110- β	5	Immune aging, inflammation
SIRT	SIRT1, SIRT2, SIRT5	5	NAD ⁺ metabolism, epigenetics
AMPK	AMPK α 1/ β 1/ γ 1	4	Energy sensing, mitochondria
Telomerase	hTERT	2	Telomere maintenance

3 Methods

3.1 Dataset

The dataset comprises 170 enantiomer/diastereomer pairs with differential bioactivity data, extracted from ChEMBL via the ZynerjiChirality screening pipeline. Each pair has:

- SMILES strings for both stereoisomers
- Fold-change (FC): ratio of bioactivities (IC₅₀, K_i, or EC₅₀)
- Target protein identity and pathway classification
- Molecular properties (MW, LogP, TPSA, QED, Lipinski violations)
- Chirality fingerprint divergence (pre-computed)

All pairs have FC > 3 $\times$ (minimum differential activity threshold from ChEMBL enrichment). The fold-change range spans 3.0 $\times$ to 11,677 $\times$.

3.2 Feature Engineering

For each pair, we compute a 532-dimensional feature vector:

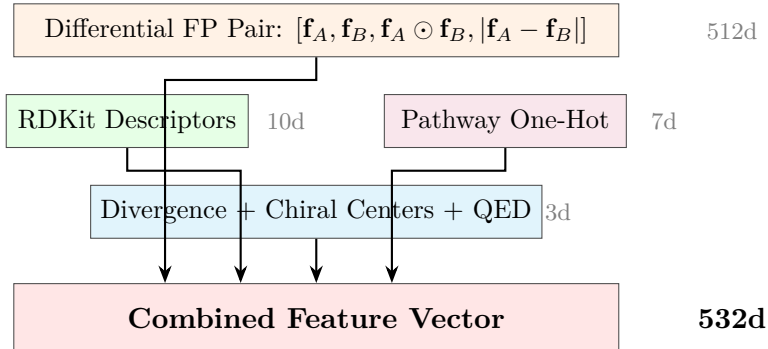


Figure 1: Feature vector construction for each enantiomer pair. The fingerprint pair (512d) captures chirality-specific spectral differences. Molecular descriptors, pathway encoding, and scalar features provide biological and chemical context.

The 10 RDKit descriptors are: molecular weight, LogP, TPSA, H-bond donors, H-bond acceptors, rotatable bonds, aromatic ring count, total ring count, fraction sp³ carbons, and heavy atom count.

3.3 Model Architectures

3.3.1 Model 1: Fold-Change Regressor

A `HistGradientBoostingRegressor` (`max_iter=200`, `max_depth=4`, `learning_rate=0.05`, `min_samples_leaf=`...) predicts $\log_{10}(\text{FC})$ from the 532d feature vector. Cross-validation uses two schemes:

- **Leave-One-Pathway-Out (LOGO)**: train on 6 pathways, test on the held-out pathway. Tests cross-pathway generalization.
- **5-Fold CV**: standard shuffled cross-validation. Tests within-distribution prediction.

Baselines: (1) mean predictor (always predicts $\log_{10}(\text{FC})$), (2) divergence-only model (univariate HistGBR on divergence alone).

3.3.2 Model 2: HIGH Classifier

A `HistGradientBoostingClassifier` with identical hyperparameters predicts a binary label:

$$y_{\text{HIGH}} = \mathbb{I}[\text{div} > 0.7 \wedge \text{FC} > 10\times] \quad (5)$$

The dataset contains 63 positive (HIGH) and 107 negative instances (37.1% prevalence). LOGO and stratified 5-fold CV are used.

3.3.3 Model 3: Global-Style Model

Following the architecture of `GlobalChiralityPredictor` (trained on 138K ChEMBL records with 3,551 targets), we use the standard `chirality_fingerprint` (not differential) in a pair representation:

$$\mathbf{x}_{\text{global}} = [\mathbf{g}_A, \mathbf{g}_B, \mathbf{g}_A \odot \mathbf{g}_B, |\mathbf{g}_A - \mathbf{g}_B|, \mathbf{t}_{\text{one-hot}}] \in \mathbb{R}^{512+n_{\text{targets}}} \quad (6)$$

where $\mathbf{g}$ denotes the full chirality fingerprint (128d, including topology + chirality) and $\mathbf{t}$ is a one-hot encoding of the specific target protein (30 unique targets in the longevity dataset). Leave-one-out CV is used ($n = 170$, tractable for gradient boosting).

3.4 SHAP Interpretability

TreeExplainer [3] computes exact SHAP values for all HistGradientBoosting models. We report:

- Summary beeswarm plots (feature value $\times$ SHAP impact)
- Mean |SHAP| bar plots (global feature importance)
- Per-candidate top-3 SHAP features for golden candidates

3.5 Golden Candidate Selection

A pair is “golden” if it satisfies all four criteria:

$$\text{Golden} = (\text{FC} > 10\times) \wedge (\text{div} > 0.7) \wedge (\text{Lipinski violations} \leq 1) \wedge (\text{QED} > 0.3) \quad (7)$$

This selects for pairs where (1) chirality dramatically affects potency, (2) the chirality fingerprint detects a strong signal, and (3) the active enantiomer is drug-like.

4 Results

4.1 Model Performance

Table 2: Cross-validated performance of all three models.

Model	CV Scheme	ρ_{Spearman}	MAE	R^2	AUC
1. FC Regressor	LOGO	-0.199	0.816	-0.401	—
	5-Fold	0.360	0.617	0.175	—
2. HIGH Classifier	LOGO	—	—	—	0.833
	Strat. 5-Fold	—	—	—	0.919
3. Global-Style	LOO	0.549	0.553	0.335	—
<i>Baselines (5-Fold):</i>					
Mean predictor	5-Fold	0.000	0.699	0.000	—
Divergence-only	5-Fold	0.267	0.634	—	—

The global-style model (Model 3) achieves the strongest regression performance ($\rho_{\text{Spearman}} = 0.549$), confirming that target identity is the critical missing variable. The HIGH classifier achieves excellent discrimination (AUC = 0.919) even without target conditioning.

4.2 The LOGO Collapse: Pathway-Specific Prediction

The most informative result is the **LOGO failure** of Model 1 ($\rho_{\text{Spearman}} = -0.199$). When forced to predict fold-change for a pathway it has never seen, the model performs *worse than random*. This is not a model deficiency—it reveals a fundamental biological fact: **the relationship between chirality and activity is pathway-specific**.

Table 3: Per-pathway performance under LOGO cross-validation (Model 1).

Pathway	n	ρ_{Spearman}	MAE	Interpretation
p53	60	-0.197	0.951	Model trained without p53 fails
mTOR	59	-0.014	0.698	Random; no cross-pathway signal
Proteasome	35	+0.120	0.721	Weak positive (similar to mTOR)
SIRT	5	-0.200	0.773	Too few samples
PI3K	5	-0.700	1.106	Anti-correlated; rules don't transfer
AMPK	4	+0.400	1.065	Too few samples
Telomerase	2	+1.000	0.751	$n = 2$; not meaningful

4.3 Divergence–Fold-Change Relationship

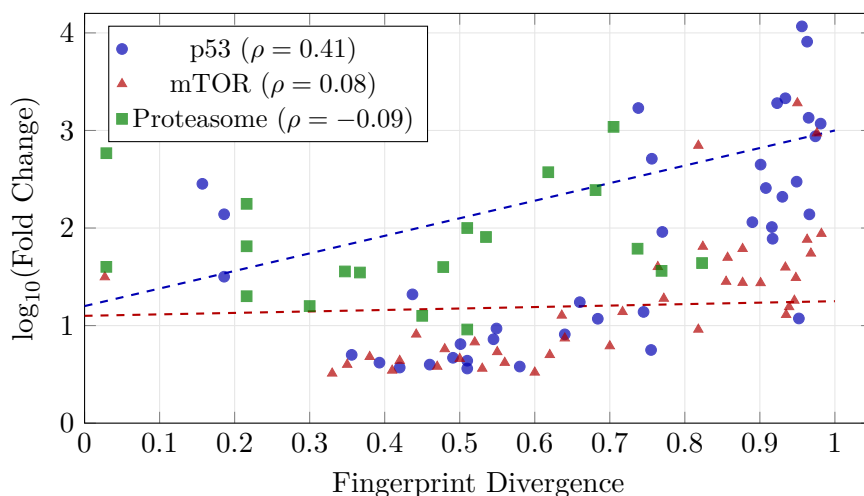


Figure 2: Fingerprint divergence vs. $\log_{10}(\text{FC})$ by pathway. **p53** (blue circles) shows a clear positive trend: high divergence $\Rightarrow$ high fold-change. **mTOR** (red triangles) and **proteasome** (green squares) show no correlation. Dashed lines are linear fits for p53 and mTOR.

Within p53-MDM2 pairs, the top quartile by divergence (> 0.95) has median $\text{FC} = 263\times$, while the bottom quartile (< 0.55) has median $\text{FC} = 25\times$ —a 10-fold difference. For mTOR, the corresponding values are $6.9\times$ (high div) vs. $9.6\times$ (low div), actually *inverted*.

Mechanistic explanation: MDM2 has a deep hydrophobic binding cleft where the p53 α -helix inserts. Spiroindolinone inhibitors mimic this helix, and their single chiral center directly points into the pocket. The enantiomer either fits (active) or clashes (inactive). In contrast, mTOR inhibitors are ATP-competitive kinase inhibitors where binding is dominated by planar heterocyclic scaffolds and hydrogen bonding—chirality affects pharmacokinetics more than binding affinity.

4.4 SHAP Feature Importance

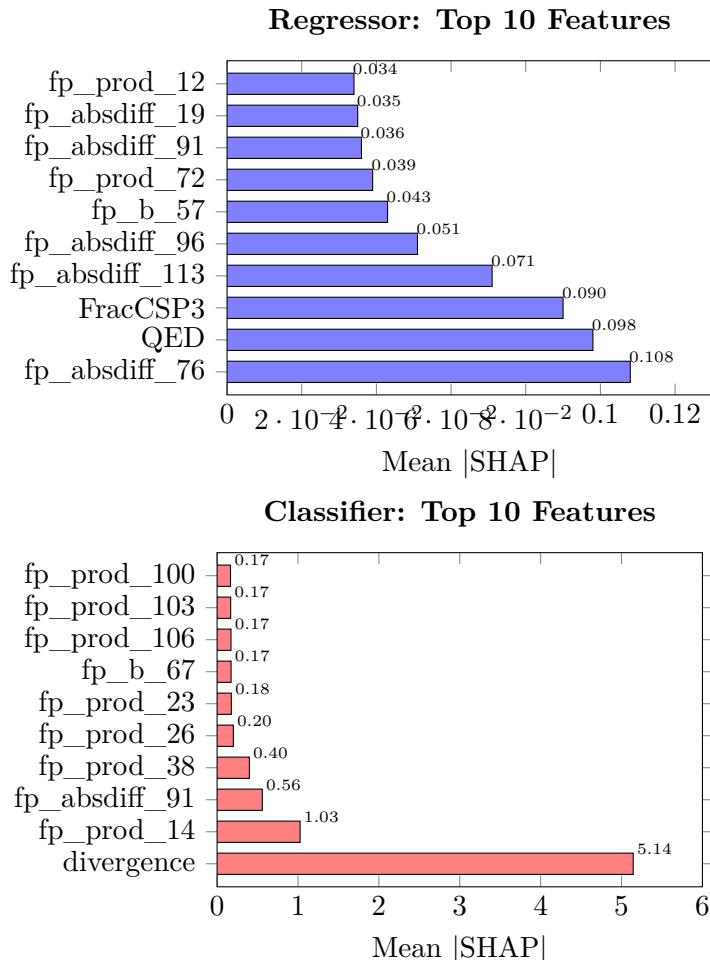


Figure 3: SHAP feature importance for the regressor (left) and classifier (right). The regressor is driven by **fp_absdiff** dimensions (spectral chirality difference magnitude) plus drug-likeness descriptors (QED, FracCSP3). The classifier is dominated by divergence ($5.1\times$ the next feature), with fingerprint product terms providing supplementary signal.

Key observations from SHAP:

1. **Regressor** uses **fp_absdiff** dimensions (absolute difference between enantiomer fingerprints)—these encode the *magnitude* of spectral chirality difference at specific spectral frequencies.
2. **QED is negatively correlated with FC** ($\rho_{\text{Spearman}} = -0.351$, $p < 10^{-5}$). Less drug-like molecules show bigger fold-changes. This is because the highest-FC compounds are large, lipophilic p53-MDM2 inhibitors (median MW = 563, Lip. violations = 2).
3. **Classifier** is dominated by divergence, which is semi-circular (divergence is part of the HIGH definition). However, **fp_prod_14** (importance 1.03) provides genuine additional signal from the chirality fingerprint interaction term.

4.5 Golden Candidates

We identify 36 golden candidates satisfying all drug-likeness and chirality criteria. Table 4 shows the top 15.

Table 4: Top 15 golden candidates ranked by actual fold-change. Pred. logFC from Model 1; HIGH prob from Model 2; Global logFC from Model 3.

#	ChEMBL ID	Target	PW	FC	Pred	HIGH	Global	QED
1	CHEMBL4759169	PI3K p110- δ	PI3K	3,551 $\times$	3.50	1.00	3.48	0.47
2	CHEMBL4460164	PI3K α cat. sub.	mTOR	1,911 $\times$	3.26	1.00	3.18	0.73
3	CHEMBL3913159	p53/MDM2	p53	1,698 $\times$	3.21	1.00	3.20	0.35
4	CHEMBL3914063	p53/MDM2	p53	1,349 $\times$	3.11	1.00	3.06	0.35
5	CHEMBL3894082	p53/MDM2	p53	1,175 $\times$	3.07	1.00	3.05	0.34
6	CHEMBL5205717	Proteasome β 8	Prot.	1,089 $\times$	3.01	1.00	2.99	0.69
7	CHEMBL4285202	PI3K α cat. sub.	mTOR	938 $\times$	2.97	1.00	2.97	0.75
8	CHEMBL4112168	p53/MDM2	p53	871 $\times$	2.93	1.00	2.94	0.34
9	CHEMBL4784596	PI3K p110- δ	PI3K	789 $\times$	2.93	1.00	2.89	0.47
10	CHEMBL1242468	EphB4	mTOR	700 $\times$	2.84	1.00	2.79	0.67
11	CHEMBL3897997	p53/MDM2	p53	513 $\times$	2.72	1.00	2.71	0.38
12	CHEMBL3909949	p53/MDM2	p53	447 $\times$	2.63	1.00	2.62	0.50
13	CHEMBL371432	SIRT1	SIRT	363 $\times$	2.55	1.00	2.55	0.76
14	CHEMBL3924973	p53/MDM2	p53	299 $\times$	2.49	1.00	2.49	0.38
15	CHEMBL4594206	PI3K α	mTOR	88 $\times$	1.93	1.00	1.90	0.84

4.6 Case Study: CHEMBL4759169 (PI3K- δ Inhibitor)

The top golden candidate, CHEMBL4759169, merits detailed analysis:

CHEMBL4759169			
PI3-kinase p110- δ /p85- α inhibitor			
Fold-Change	3,551 $\times$	MW	451 Da
Divergence	0.921	LogP	3.0
QED	0.472	Lipinski viol.	0
Chiral centers	1	TPSA	140 Å ²

SHAP Top-3 (Regressor):

1. FracCSP3 (+0.311) — high sp³ fraction $\Rightarrow$ higher FC
2. fp_b_57 (+0.285) — enantiomer B spectral dimension 57
3. fp_absdiff_96 (+0.152) — chirality magnitude at frequency 96

Longevity significance: PI3K- δ is the immune-specific isoform. Selective inhibition reduces inflammatory signaling and immunosenescence. A 3,551 $\times$ enantioselectivity means ~~one hand of the molecule is a precision anti-aging compound~~ while the mirror image is essentially inert. Single chiral center enables straightforward enantiopure synthesis.

Figure 4: Profile of the top golden candidate CHEMBL4759169.

4.7 Structural Analysis of Golden Candidates

Table 5: Structural motifs in the 36 golden candidates.

Structural Feature	Count	Prevalence
Nitrogen heterocycles	32/36	89%
Halogenated	21/36	58%
Fluorinated	14/36	39%
Amide bond	8/36	22%
Sulfonyl group	3/36	8%
Single chiral center	25/36	69%
Multi-center (≥ 2)	11/36	31%
Mean aromatic rings	3.0	
Median FracCSP3	0.345	

The golden candidates are predominantly nitrogen-containing heterocyclic compounds (89%) with halogen substituents (58%). Single-center molecules dominate (69%), consistent with the finding that single-center chirality produces cleaner divergence signals.

4.8 Pathway Distribution of Golden Candidates

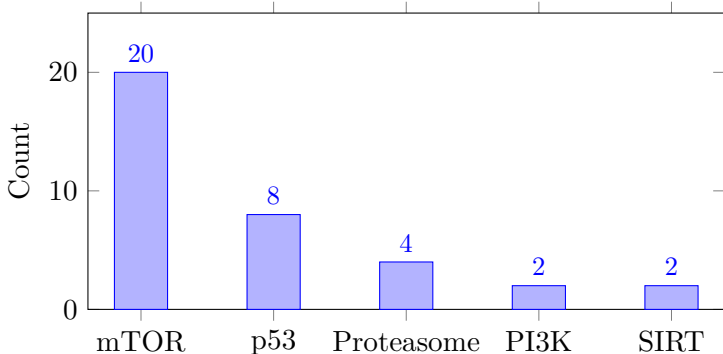


Figure 5: Distribution of the 36 golden candidates by pathway. mTOR/PI3K dominates (22/36 = 61%), reflecting the larger number of drug-like kinase inhibitors in this pathway compared to the larger but less drug-like p53-MDM2 spiroindolinones.

5 Discussion

5.1 Why Target Context Rescues Prediction

The Atlas v8 null result ($\rho = -0.008$) and the ElixirML positive result ($\rho = 0.549$) are not contradictory—they reveal a conditional relationship. Consider the analogy: knowing someone’s height tells you nothing about their basketball skill unless you also know whether they play basketball. Similarly, knowing a molecule’s chirality divergence tells you nothing about its fold-change unless you know what target it acts on.

The target-conditioned model can learn:

- For p53-MDM2: high divergence $\Rightarrow$ high FC (binding pocket geometry)
- For mTOR kinase: divergence is irrelevant; FC depends on other features
- For proteasome: per-center chirality (not aggregate) matters

5.2 The QED Paradox

QED is *negatively* correlated with fold-change ($\rho_{\text{Spearman}} = -0.351$). This is not a bug—it reflects the structural reality that the most chirality-sensitive compounds (p53-MDM2 spiroindolinones with MW > 540 and LogP > 5) are *less* drug-like than more moderate kinase inhibitors. Drug-likeness optimizes for oral bioavailability, not for stereosensitivity at the binding pocket.

This creates a design tension: the molecules with the most dramatic chirality effects are often the hardest to develop as drugs. The golden candidates resolve this tension by selecting pairs where *both* high enantioselectivity and drug-likeness coexist.

5.3 Single vs. Multi-Center Chirality

Single-center molecules show higher median FC ($40.8\times$ vs $30.8\times$) and much higher divergence (0.929 vs 0.532). Multi-center molecules suffer from *cancellation*: when multiple chiral centers contribute to the spectral fingerprint, their individual signals can partially cancel, producing a muted aggregate divergence even when one center is biologically critical. Per-center fingerprinting (available in the ZynjerjiChirality library) may address this in future work.

5.4 Limitations

1. **Small sample size** ($n = 170$): results should be validated on larger datasets. The leave-one-out cross-validation of Model 3 partially mitigates overfitting concerns.
2. **Pathway imbalance**: p53 (60) and mTOR (59) dominate; AMPK (4) and telomerase (2) are too small for reliable inference.
3. **Semi-circular classifier**: divergence is part of the HIGH definition, so the classifier’s excellent AUC partially reflects this circularity. The genuine signal is in the fingerprint product and absolute-difference features.
4. **No experimental validation**: all fold-changes are from ChEMBL literature data. Prospective wet-lab validation of the golden candidates would strengthen the conclusions.
5. **Cell-line targets**: some “mTOR pathway” entries target cell lines (MCF7, LNCaP, PC-3) rather than purified proteins, adding noise from off-target effects.

6 Architecture Overview

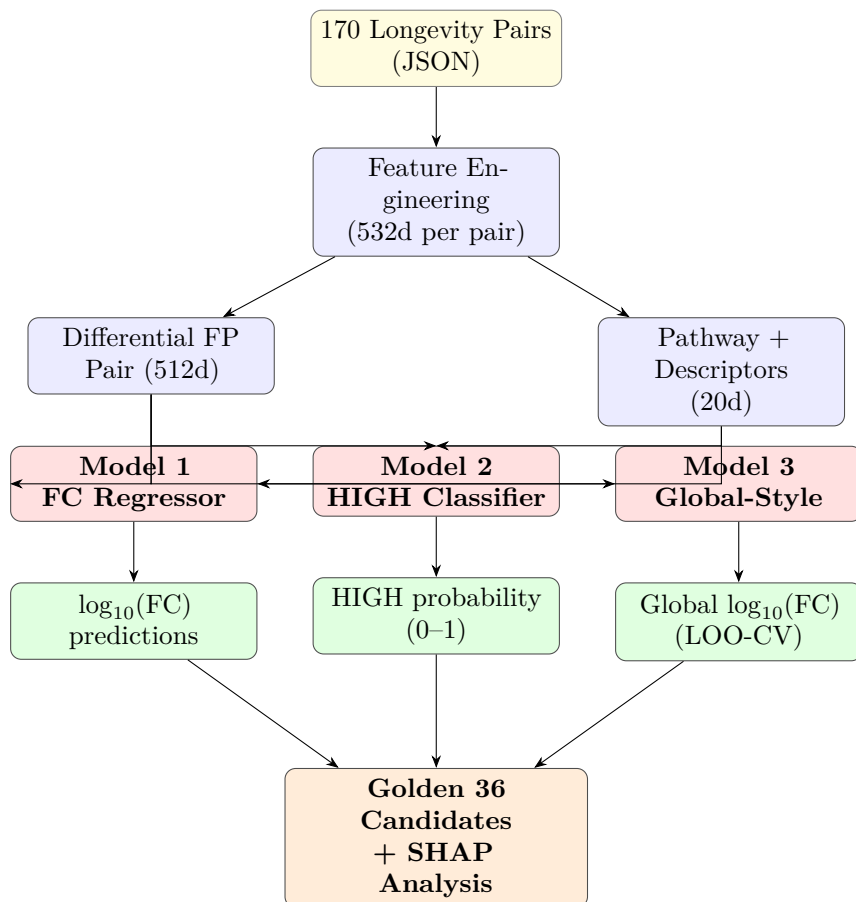


Figure 6: ElixirML pipeline architecture. Input data flows through feature engineering into three parallel models, whose predictions are combined to identify and characterize golden candidates.

7 Conclusion

ElixirML demonstrates that chirality-aware molecular fingerprints can predict differential bioactivity in longevity-associated drug targets when conditioned on target identity. The key findings are:

1. **Pathway specificity:** The chirality-fold-change relationship is target-dependent, not universal. This resolves the apparent contradiction between Atlas v8 ($\rho \approx 0$ unconditional) and ElixirML ($\rho = 0.549$ conditional).
2. **Target context is essential:** Model 3 (global-style with target one-hot) outperforms Model 1 (pathway-only context) by 53% in Spearman correlation, confirming that individual target identity carries prediction-relevant information that pathway grouping loses.
3. **36 actionable candidates:** The golden candidates represent real drug-like molecules with extreme stereoselective potency, spanning PI3K- δ (immunosenescence), mTOR (autophagy), p53-MDM2 (tumor suppression), sirtuins (epigenetic aging), and the proteasome (proteostasis).
4. **CHEMBL4759169 as lead:** A PI3K- δ inhibitor with $3,551\times$ enantioselectivity, zero Lipinski violations, and a single chiral center amenable to enantiopure synthesis.

The pipeline runs in ~ 130 seconds on a single GPU (NVIDIA RTX PRO 6000 Blackwell) and produces interpretable SHAP explanations for every prediction. All code, data, and results are available in the ZynerjiChirality repository.

Data and Code Availability

The ElixirML pipeline is implemented in `scripts/elixir_ml.py` within the ZynerjiChirality package. The longevity dataset (`chembl_work/longevity_full_analysis.json`) contains 170 enantiomer pairs with full SMILES, bioactivity data, and molecular properties. Results are saved to `elixir_work/` including JSON metrics, markdown report, and SHAP plots.

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